TMM4128:
Machine Learning for Engineers

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- Dimensions of the course
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  - Engineering disciplines
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- Attendance of the course

- ML project checklist
- Homework

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TMM4128 (2025): Top view

- 14 weeks in Spring semester.
- 2 x 2h lectures, 1 x 2h seminar per week. (6 contact hours per week)

"The grade is divided into 40% for portfolio assessment and 60% for final exams."

Learning outcome

Having successfully completed this course student will be able to acquire the following:

Knowledge:

- Learn the fundamental principles of supervised, unsupervised and reinforcement learning.
- Acquiring knowledge of using ML to solve practical problems relevant for engineers.

Skills:

- Apply data handling, feature engineering and data pre-processing techniques
- Gain experience to systematically work with data to learn new patterns.

General competence:

- At the end of this course students will understand the strengths and limitations of well-known machine learning methods, and learn how to analyse data to identify trends.

Machine Learning for Engineers

Why?

Why?

- Many of human activities get “digital traces”, including that for engineers.
  - Models are created, data being generated during an engineering process
- A: How can we use these “digital traces”(models and data) for helping to solve engineering problems?

- B: Is it possible to get ML for engineers as off-the-shelf tools? - Where and to what extent?
  - **Having an engineering problem, what ML tools should I select and how should I finetune those for the given problem?**

Machine Learning for Engineers

Dimensions shaping the course

Dimensions

- ‘Types’ of Machine Learning
- Engineering disciplines
- Methodological (aka “Data science”)

Types of ML

- Supervised learning
- Unsupervised learning
- Reinforcement learning

Engineering disciplines

- Mechanical engineering
- Industrial engineering
- Materials engineering
- Civil engineering
- Computer engineering
- Electrical engineering
- Aerospace engineering
- Process engineering
- Chemical engineering
- Environmental engineering
- Biomedical engineering
- … (and many more)

Subject area(s)

Machine Design and Materials Technology - Materials Production Processes
Machine Design and Materials Technology - Mechanical Integrity
Machine Design and Matherials Technology - Products and Machine Design
Computer and Information Science
ICT and Mathematics
Machine Design and Materials Technology
Computers
Machine Design and Materials Technology - Mechanical Integrity in Machine Design
Computer Systems
Computer Systems

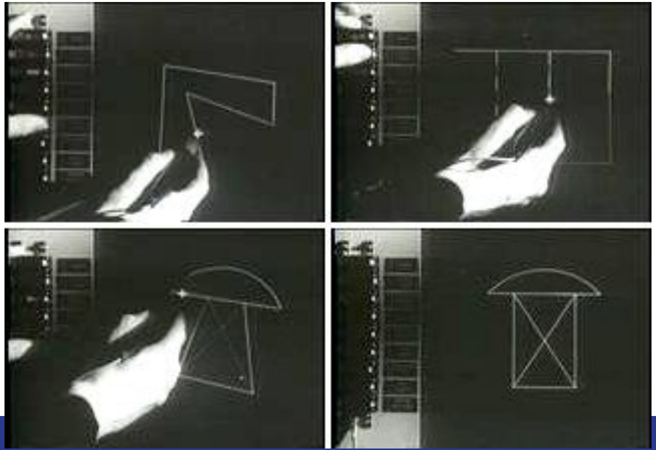
Product development lifecycle

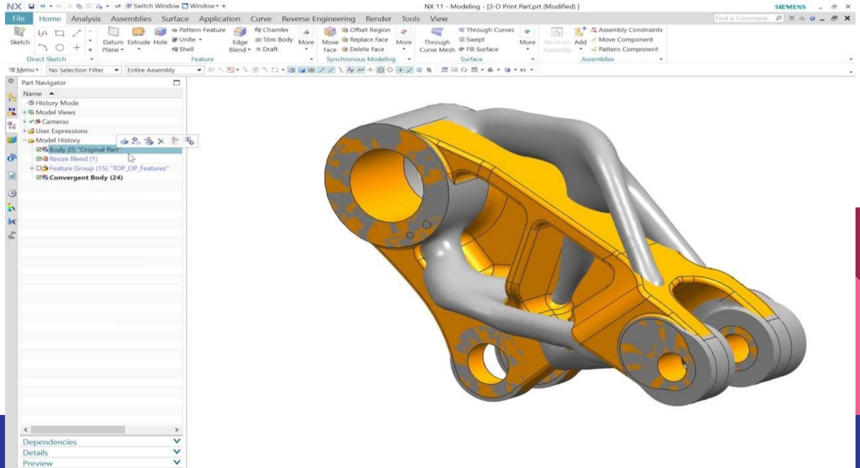
Computer-aided Design (CAD) - Background (1)

- Computer-aided design (CAD) utilizes computers to create and edit design models and drawings.
- Applications:
  - Computer animation of movies
  - Architectural design
  - Electric and electronic circuit design
  - Garment design
  - Mechanical system design
  - …

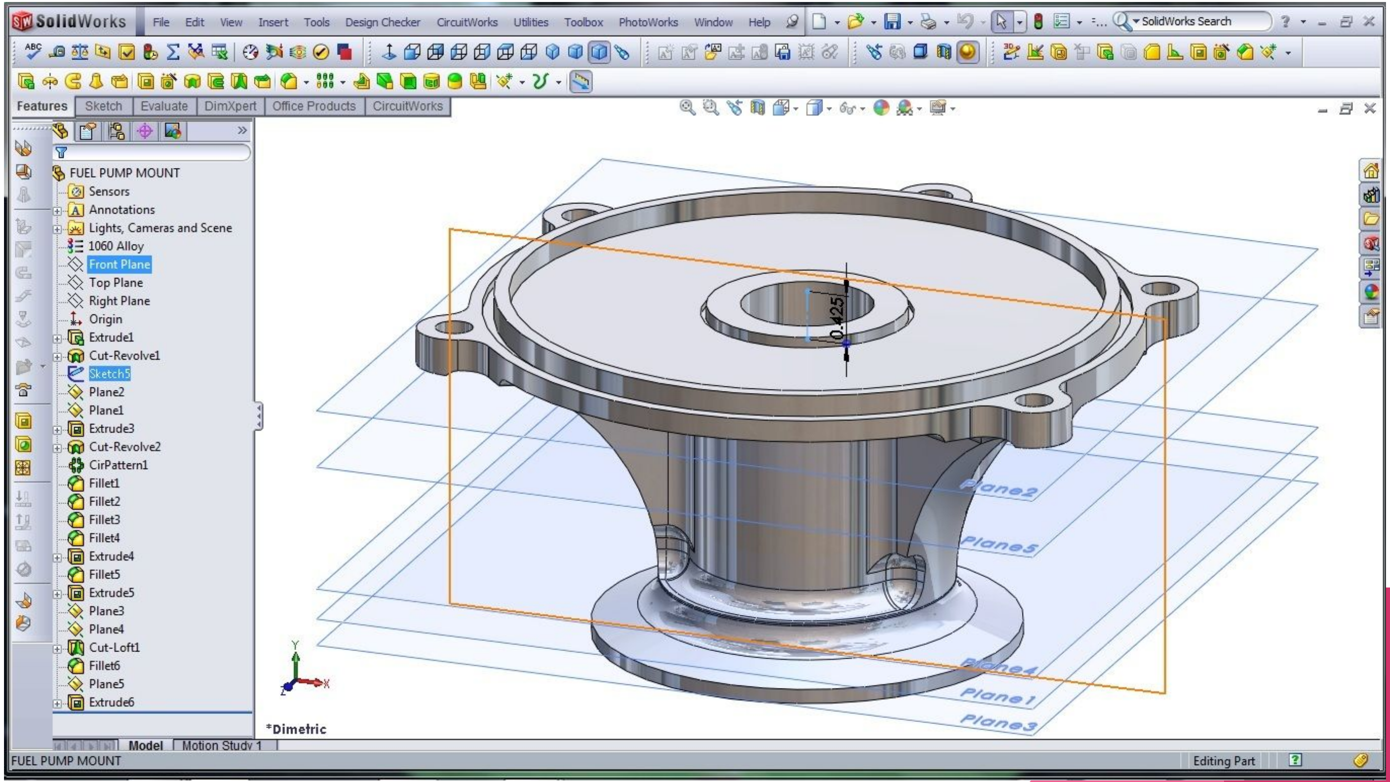
Computer-aided Design (CAD) - Background (2)

- CAD used to model product geometries.
- May record also some non-geometric design specs.
- Design model mat serve as the interface between design and manufacturing.
- CAD is critical for Computer-aided manufacturing (CAM).
- Development of CAD began in 1960s ...





So, one can design a product (CAD)



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Web-based framework: Onshape CAD **online**

Everything you need is a web browser. More and more of functions and data become easily accessible.

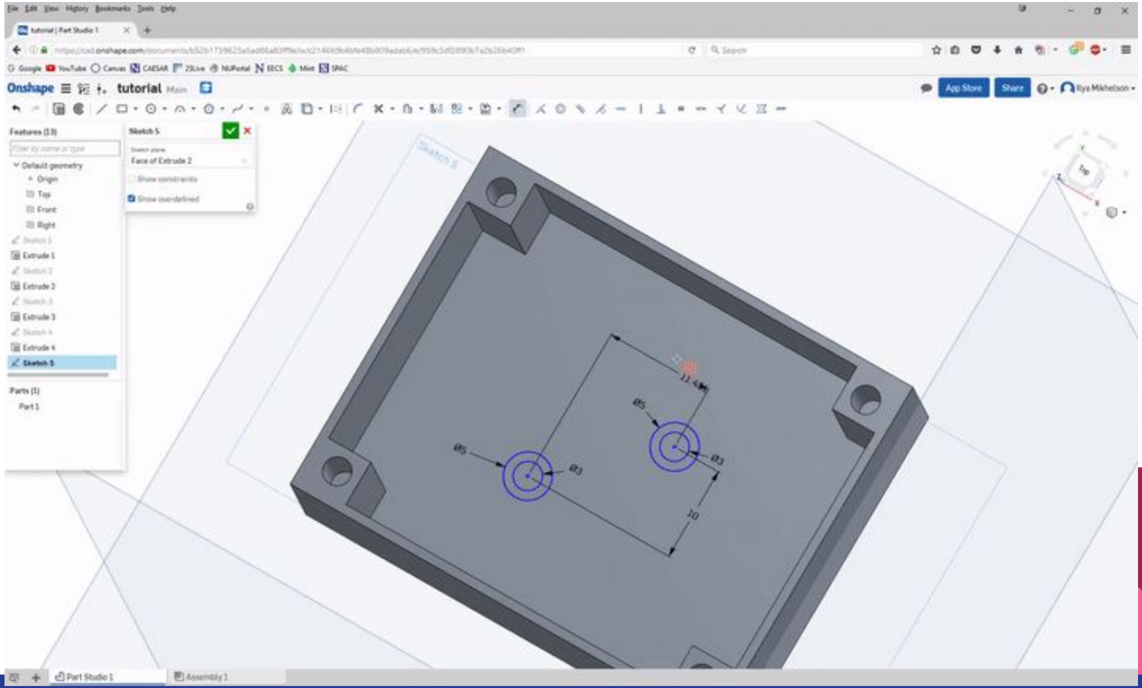
(REST) API available, e.g:

https://cad.onshape.com/api/documents/d/:did/[wv]/:wv/e/:eid/export

did: documentId

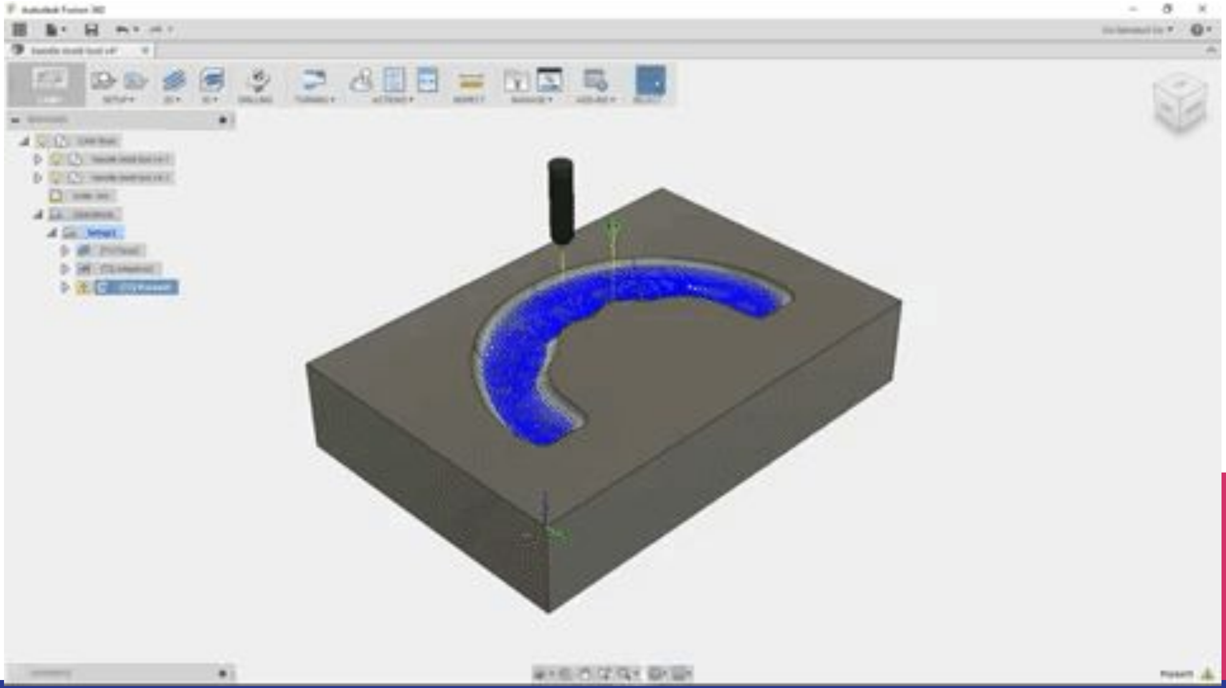
wv: workspaceId

...

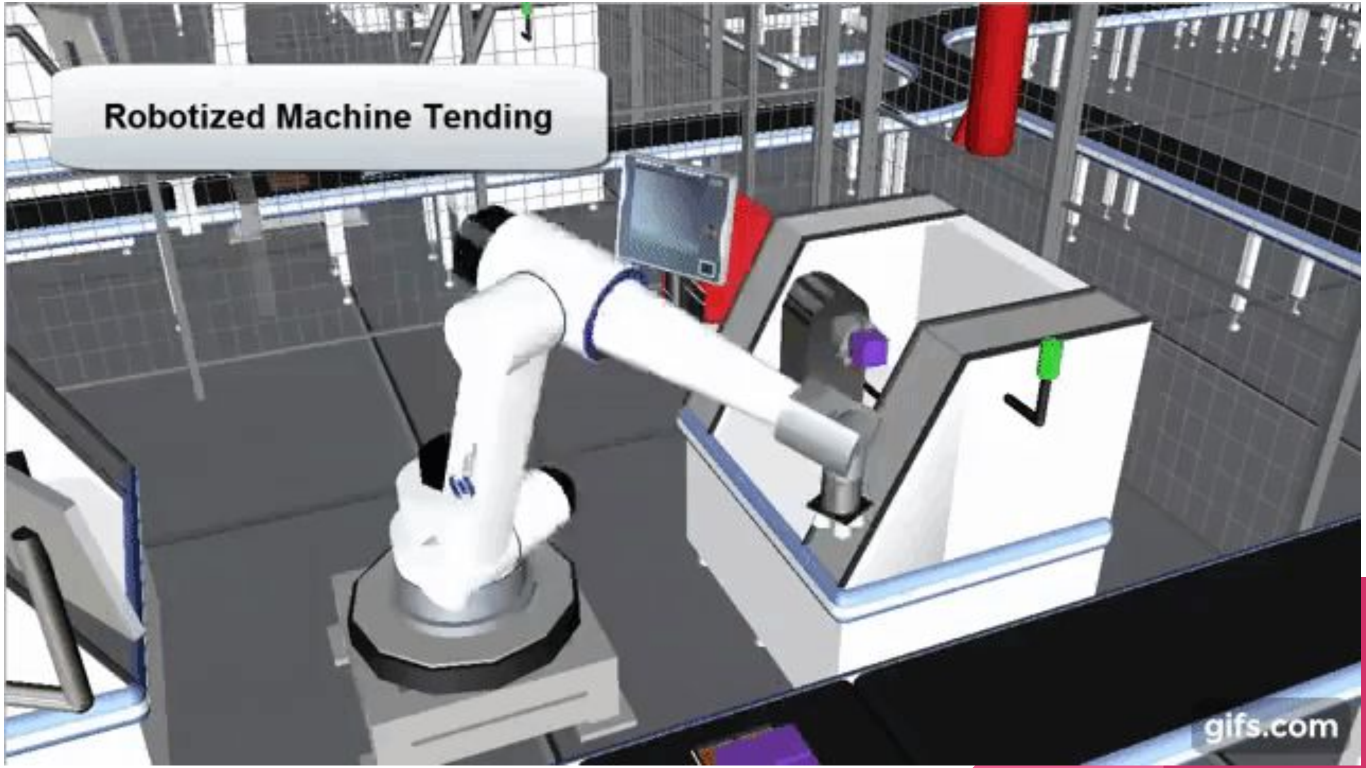


Computer-aided Manufacturing (CAM)

Analysing manufacturing of workpieces and operations of production plants.



One can design a process/production system (CAM)

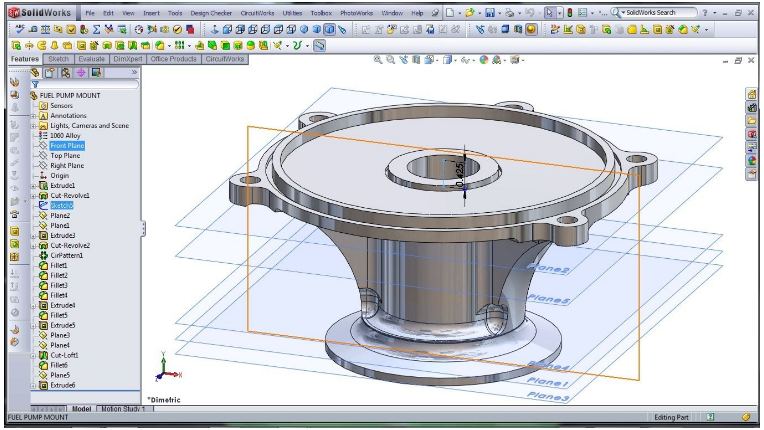


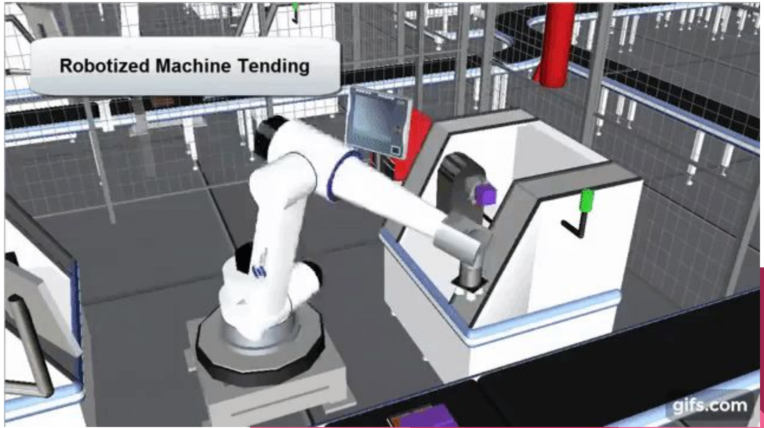
One can continuously integrate product and manufacturing system design

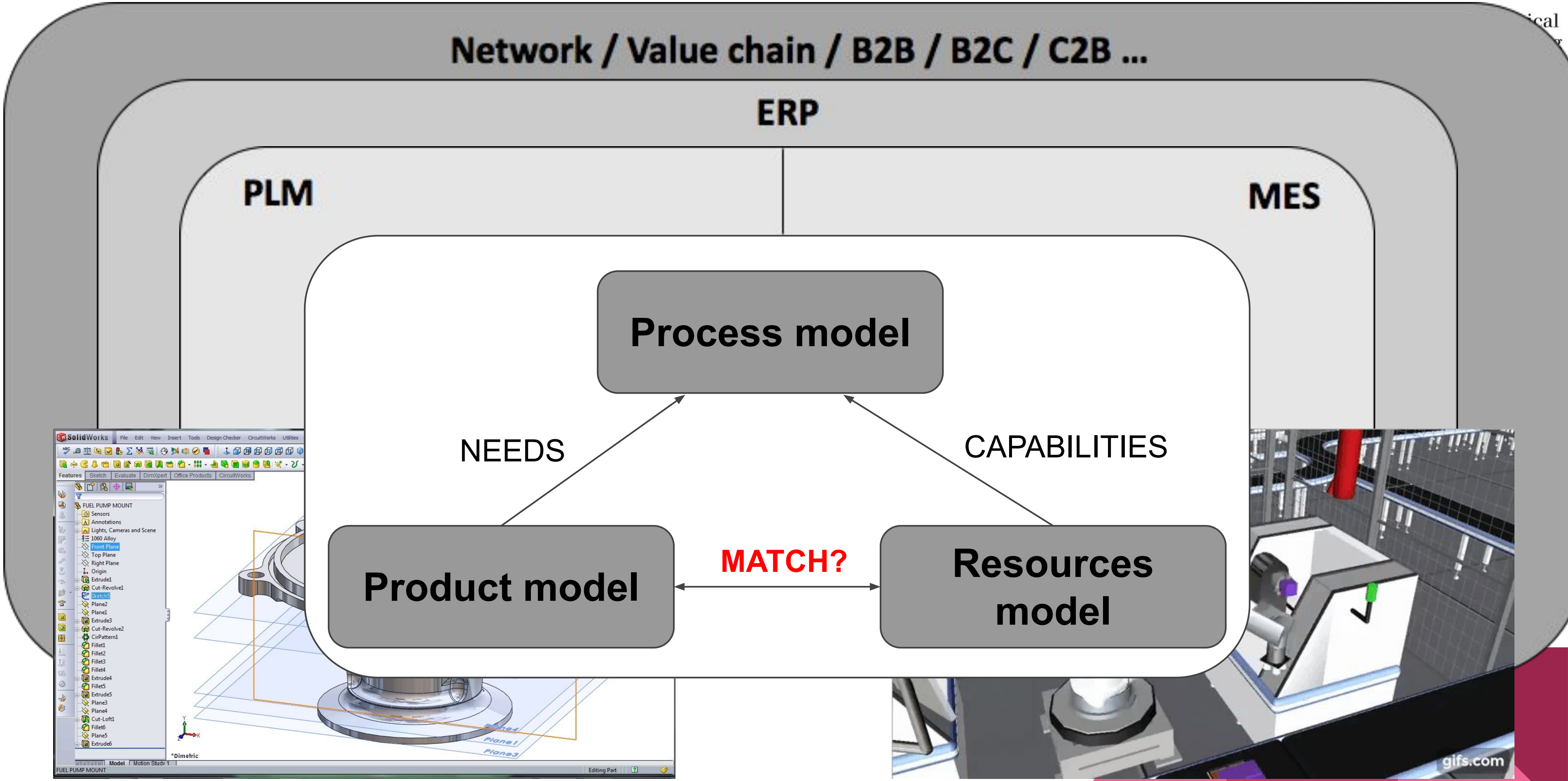
Check the properties of the product / part.

Check the performance of a manufacturing system.

Design for ‘X’ (assembly, quality, resourcefulness, ...)







Learn and improve over the time?

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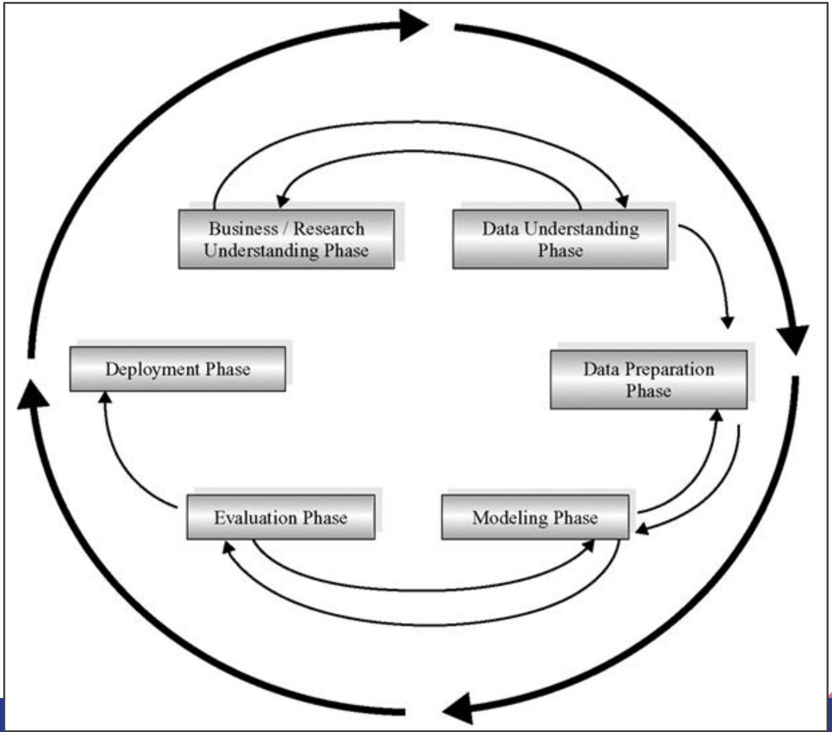
Methodological view / data science

CRoss Industry Standard Process for Data Mining (CRISP-DM) is an industry- and tool-neutral process for data mining (CRISP-DM, 2007).



ML as an off-the-shelf tool?

Some initiatives / projects / platforms / examples

AI4EU list of projects

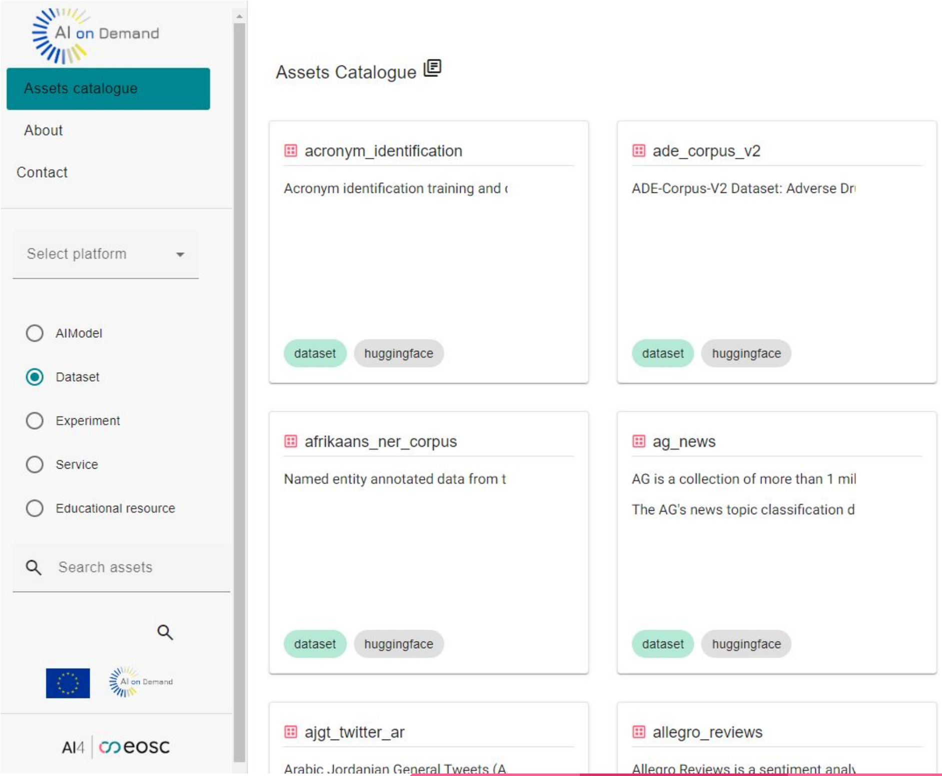
https://www.ai4europe.eu/ai-community/projects

https://www.ai4europe.eu/research/ai-catalog

AI-on-Demand

https://aiod.eu/about

https://mylibrary.aiod.eu/marketplace

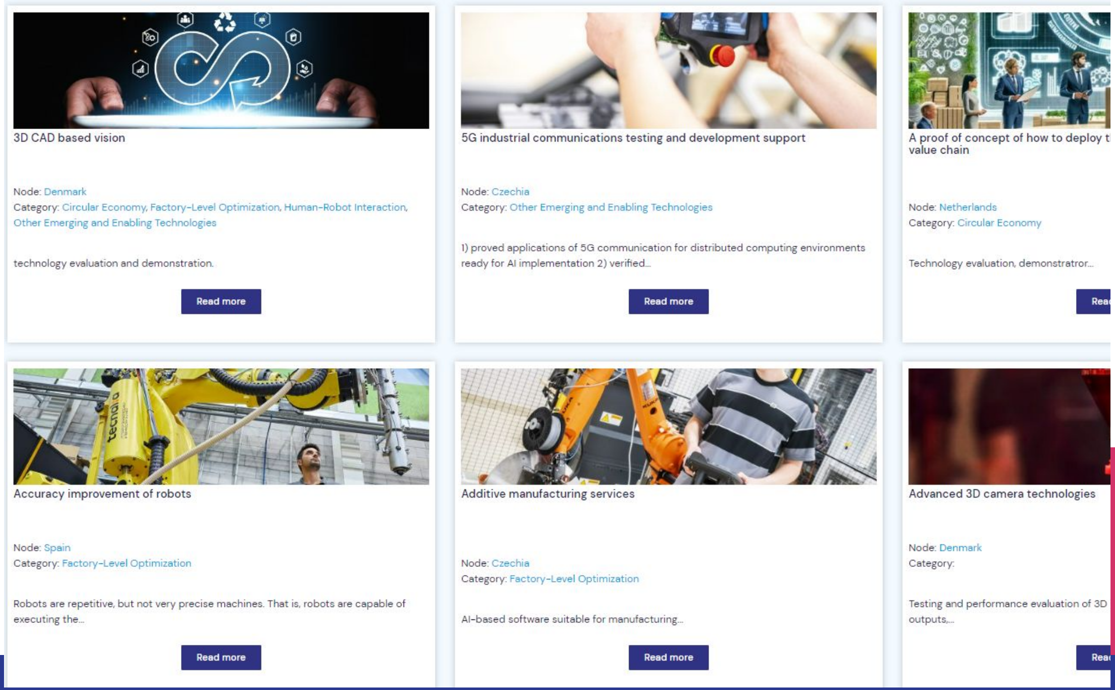


AI-Matters

Services

AI-MATTERS offers its customers an extensive service catalogue spanning the topics above that evolves through continuous updates as needs and expectations of the European manufacturing industry progress.

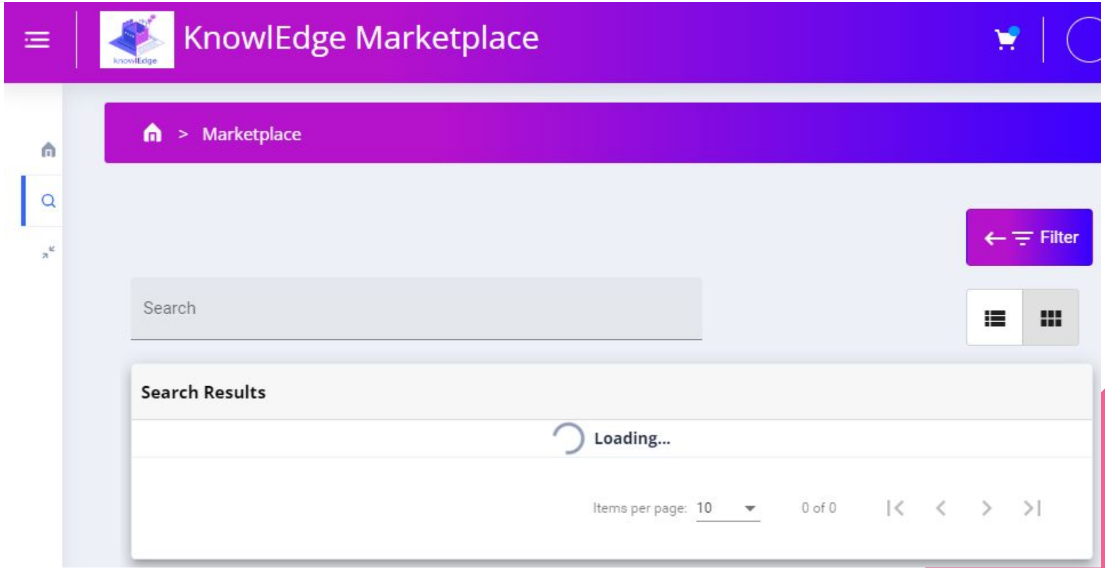
https://ai-matters.eu/services-catalog/



knowlEdge project (ended 03.24)

knowlEdge Project
AI for manufacturing

https://www.knowledge-project.eu/about-knowledge/resources/



AI REGIO

https://www.airegio-project.eu/experiments

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Takeaway on ML as an off-the-shelf tool

- There are some ongoing attempts to create “marketplaces” for sharing data / tools on industry-relevant ML applications. **The quality is to be assessed**.

- In TMM4128, we try a similar approach between the student groups working on the course assignments tailored for application of selected methods for different problem domains (engineering disciplines).

- *> Let’s see what tools and **how** we can bring those to the “marketplace”.*

TMM4128 (2025): Table of Contents

(Tentative)

TMM4128: Table of Contents

The course has 3 main arbitrary phases or dimensions.
Phase I: **Product design. *Supervised learning*.**
Phase II **Manufacturing / Operations / Maintenance**. ***Unsupervised learning***.
Phase III **End-of-life, reuse, repurposing, recycling**. ***Semi-supervised, reinforcement learning***.

Note: From a product life cycle perspective, selected phases are i) product design; ii) manufacturing / operations / maintenance; and iii) end-of-life, reuse, repurposing, recycling. From the learning paradigm, the focus is on i) supervised learning; ii) unsupervised learning; and iii) semi-supervised, reinforcement learning. The given alignment of the product life cycle perspective with the ML paradigms is arbitrary, as one can have different ML methods under given paradigms applied to relevant problems at the different phases. However, the intention here is to move from more “simple” cases (having data) towards the cases with greater “uncertainty” that may require integrative and/or online approaches AND to keep an overview of a bigger picture (i.e. PLM).

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TMM4128 (2025): ToC (1)

Thu., 16.01., 8-10: Course housekeeping / Intro to ML / Machine Learning project checklist
Fri., 17.01., 12-14: ML tools, setting up the development environment
Phase I: **Product design. *Supervised learning*.**
Thu., 23.01., 8-10: Classification [A1 announcement]
Fri., 24.01., 12-14: Training models (1)
Thu., 30.01., 8-10: Training models (2)
Thu., 30.01., 10-12: A1/First iteration on ML projects (Seminar)
Fri., 31.01., 12-14: Support Vector Machines (SVM)
Thu., 06.02., 8-10: Decision trees
Thu., 06.02., 10-12: A1/Second iteration on ML projects (Seminar)
Fri., 07.02., 12-14: Ensemble learning and random forests

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TMM4128 (2025): ToC (2)

Thu., 13.02., 8-10: Dimensionality reduction
Thu., 13.02., 10-12: A1/Third iteration on ML projects (Seminar)
Fri., 14.02., 12-14: Recap: supervised learning to support product design / development
Phase II **Manufacturing / Operations / Maintenance**. ***Unsupervised learning***.
Thu., 20.02., 8-10: Unsupervised learning techniques - Clustering
Thu., 20.02., 10-12: A1/Projects presentations [*A1 deadline*] (Seminar)
Fri., 21.02., 12-14: Unsupervised learning techniques - Gaussian mixtures [A2 announcement]
Thu., 27.02., 8-10: Artificial Neural Networks, Intro (1)
Thu., 27.02., 10-12: A2/First iteration on ML projects (Seminar)
Fri., 28.02., 12-14: Artificial Neural Networks, Intro (1)

TMM4128 (2025): ToC (3)

TMM4128 (2025): ToC (4)

Phase III **End-of-life, recycling, repurposing**. ***Semi-supervised, reinforcement learning***.
Thu., 03.04., 8-10: Convolutional Neural Networks (1)
Thu., 03.04., 10-12: A3/First iteration on ML projects (Seminar)
Fri., 04.04., 12-14: Convolutional Neural Networks (2)
Thu., 10.04., 8-10: Reinforcement learning (1)
Thu., 10.04., 10-12: A3/Second iteration on ML projects (Seminar)
Fri., 11.04., 12-14: Reinforcement learning (2)
Thu., 24.04., 8-10: Reinforcement learning (3)
Thu., 24.04., 10-12: A3/Third iteration on ML projects (Seminar)
Fri., 25.04., 12-14: Recap: Semi-supervised and reinforcement learning to learn from and throughout product life cycle.
Fri., 02.05., 12-14: A3/Projects presentations [*A3 deadline*]

Locations

Thursdays lectures - GL-GF F6

Thursdays seminars - L-Big Ben

Friday lectures - GL-GEL EL1

F6

Gamle fysikk, Floor 2

L-Big Ben (dataarb)

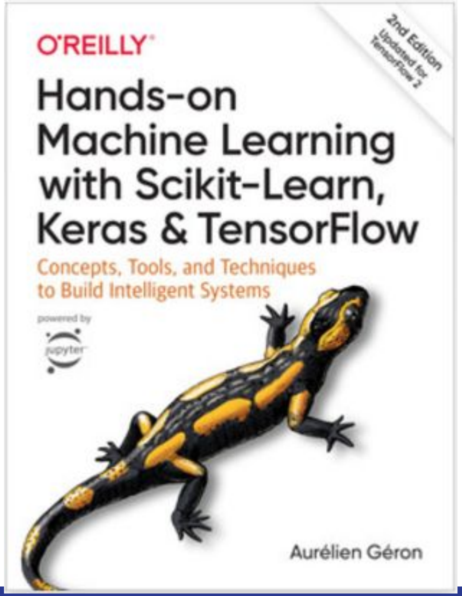
Byggtekniske laboratorier, Floor 2

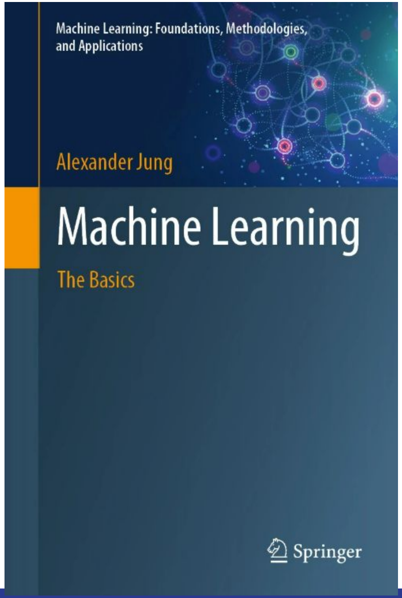
EL1

Gamle elektro, Floor 2

Books / Main sources

Aurelien Geron, "Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow : concepts, tools, and techniques to build intelligent systems", 2019 (Available online);
Alexander Jung, "Machine Learning: The Basics", 2022 (Available online)





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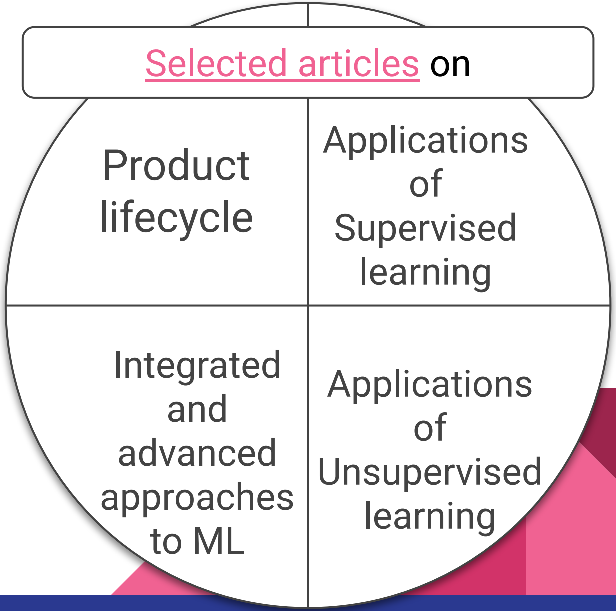

Assignments 1, 2, 3

A1: Do we learn from previous experience when building a new product?
An example: Validating technical drawings.

A2: Do we operate a machine efficiently?
An example: Anomaly detection for the bearings of rotating machinery.

A3: Do we reuse, repurpose or recycle a product efficiently?
An example: Optimisation of garbage collection and processing.

Assignments to be done in groups of 2-3 students. Group specific subjects to be created for each assignment.

Examples of potential datasets

https://www.kaggle.com/datasets/alexandrparkhomenko/the-fasteners
https://github.com/Azure-Samples/digitization-of-piping-and-instrument-diagrams/blob/main/docs/architecture.md
https://www.simaud.org/datasets/
https://www.kaggle.com/datasets?search=vibration&tags=12300-Engineering
https://www.kaggle.com/datasets?search=recycling&tags=12300-Engineering

...

Teaching assistants (TAs)

1. Sunil Sharma, sunils@stud.ntnu.no

2. Hooman Hooshmand, hoomanh@stud.ntnu.no

To support students with use of ML tools, methods, APIs.

Weekly time slots with TAs will be shortly announced.

Attendance of the course

- If you have overlapping of the courses, please, decide and select just one course to dedicate your full attention. (Psst! On hard choices). Select and go for one course, in case of overlaps.
  - Also think about some common activities, e.g. assignments, where the lectures are also good, natural sync point with your group mates to go through the same experiences.

Questions?

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ML project checklist

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Outline : ML project checklist

- 8 steps of ML project
  a. Frame the problem
  b. Get the data
  c. Explore the data
  d. Prepare the data
  e. Shortlist promising models
  f. Fine-tune the system
  g. Present your solution
  h. Launch, monitor, and maintain

—

- **Source**: Aurelien Geron, “Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow : concepts, tools, and techniques to build intelligent systems”, 2019 (Available online) / **Appendix B**

1. Frame the problem

1. Define the objective in business terms.
2. How will your solution be used?
3. What are the current solutions/workarounds (if any)?
4. How should you frame this problem (supervised/unsupervised, online/offline, etc.)?
5. How should performance be measured?
6. Is the performance measure aligned with the business objective?
7. What would be the minimum performance needed to reach the business objective?
8. What are comparable problems? Can you reuse experience or tools?
9. Is human expertise available?
10. How would you solve the problem manually?
11. List the assumptions you (or others) have made so far.
12. Verify assumptions if possible.

2. Get the data

1. List the data you need and how much you need.
2. Find and document where you can get that data.
3. Check how much space it will take.
4. Check legal obligations, and get authorization if necessary.
5. Get access authorizations.
6. Create a workspace (with enough storage space).
7. Get the data.
8. Convert the data to a format you can easily manipulate (without changing the data itself).
9. Ensure sensitive information is deleted or protected (e.g., anonymized).
10. Check the size and type of data (time series, sample, geographical, etc.).
11. Sample a test set, put it aside, and never look at it (no data snooping!).

3. Explore the data

1. Create a copy of the data for exploration (sampling it down to a manageable size if necessary).
2. Create a Jupyter notebook to keep a record of your data exploration.
3. Study each attribute and its characteristics:
    a. Name
    b. Type (categorical, int/float, bounded/unbounded, text, structured, etc.)
    c. % of missing values
    d. Noisiness and type of noise (stochastic, outliers, rounding errors, etc.)
    e. Usefulness for the task
    f. Type of distribution (Gaussian, uniform, logarithmic, etc.)
4. For supervised learning tasks, identify the target attribute(s).
5. Visualize the data.
6. Study the correlations between attributes.
7. Study how you would solve the problem manually.
8. Identify the promising transformations you may want to apply.
9. Identify extra data that would be useful (see “Get the Data”).
10. Document what you have learned.

4. Prepare the data (1)

- Work on copies of the data (keep the original dataset intact).
- Write functions for all data transformations you apply, for five reasons:
    - So you can easily prepare the data the next time you get a fresh dataset
    - So you can apply these transformations in future projects
    - To clean and prepare the test set
    - To clean and prepare new data instances once your solution is live
    - To make it easy to treat your preparation choices as hyperparameters

4. Prepare the data (2)

1. Data cleaning:
   - Fix or remove outliers (optional).
   - Fill in missing values (e.g., with zero, mean, median…) or drop their rows (or columns).
2. Feature selection (optional):
   - Drop the attributes that provide no useful information for the task.
3. Feature engineering, where appropriate:
   - Discretize continuous features.
   - Decompose features (e.g., categorical, date/time, etc.).
   - Add promising transformations of features (e.g., log(x), sqrt(x), x 2 , etc.).
   - Aggregate features into promising new features.
4. Feature scaling:
   - Standardize or normalize features.

5. Shortlist promising models

1. Train many quick-and-dirty models from different categories (e.g., linear, naive Bayes, SVM, Random Forest, neural net, etc.) using standard parameters.
2. Measure and compare their performance.
    a. For each model, use N-fold cross-validation and compute the mean and standard deviation of the performance measure on the N folds.
3. Analyze the most significant variables for each algorithm.
4. Analyze the types of errors the models make.
    a. What data would a human have used to avoid these errors?
5. Perform a quick round of feature selection and engineering.
6. Perform one or two more quick iterations of the five previous steps.
7. Shortlist the top three to five most promising models, preferring models that make different types of errors.

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6. Fine-tune the system

1. Fine-tune the hyperparameters using cross-validation:
   a. Treat your data transformation choices as hyperparameters, especially when you are not sure about them (e.g., if you’re not sure whether to replace missing values with zeros or with the median value, or to just drop the rows).
   b. Unless there are very few hyperparameter values to explore, prefer random search over grid search. If training is very long, you may prefer a Bayesian optimization approach (e.g., using Gaussian process priors, as described by Jasper Snoek et al.).
2. Try Ensemble methods. Combining your best models will often produce better performance than running them individually.
3. Once you are confident about your final model, measure its performance on the test set to estimate the generalization error.l, measure its performance on the test set to estimate the generalization error.

7. Present your solution

1. Document what you have done.
2. Create a nice presentation.
   a. Make sure you highlight the big picture first.
3. Explain why your solution achieves the business objective.
4. Don’t forget to present interesting points you noticed along the way.
   a. Describe what worked and what did not.
   b. List your assumptions and your system’s limitations.
5. Ensure your key findings are communicated through beautiful visualizations or easy-to-remember statements (e.g., “the median income is the number-one predictor of housing prices”).

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8. Launch!

1. Get your solution ready for production (plug into production data inputs, write unit tests, etc.).
2. Write monitoring code to check your system's live performance at regular intervals and trigger alerts when it drops.
   a. Beware of slow degradation: models tend to "rot" as data evolves.
   b. Measuring performance may require a human pipeline (e.g., via a crowdsourcing service).
   c. Also monitor your inputs' quality (e.g., a malfunctioning sensor sending random values, or another team's output becoming stale). This is particularly important for online learning systems.
3. Retrain your models on a regular basis on fresh data (automate as much as possible).

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Homework

1. Check Google scholar for scientific publications on application of **machine learning in product design / development** (a search prompt example). *You may need to be on NTNU network (i.e. either on campus or via the VPN) to have access to most of the articles.*
2. For the article selected, check:
   a. for the “objective in business terms” - What and why they try to solve / improve?
   b. to what extent the article intersects with the presented ML project checklist? - Can you identify any steps?
   c. what ML method is used?
   d. what ML tools / “frameworks” or toolchains are in use?
   e. what is a data structure?

Format: **Max 1 A4** page PDF or DOCX answering questions above.
DL: **22.01., 12:00** to be submitted via Blackboard.
Purpose: Developing skills of checking and presenting the key elements and process for ML application (which also suppose to support your work later for the course assignments).